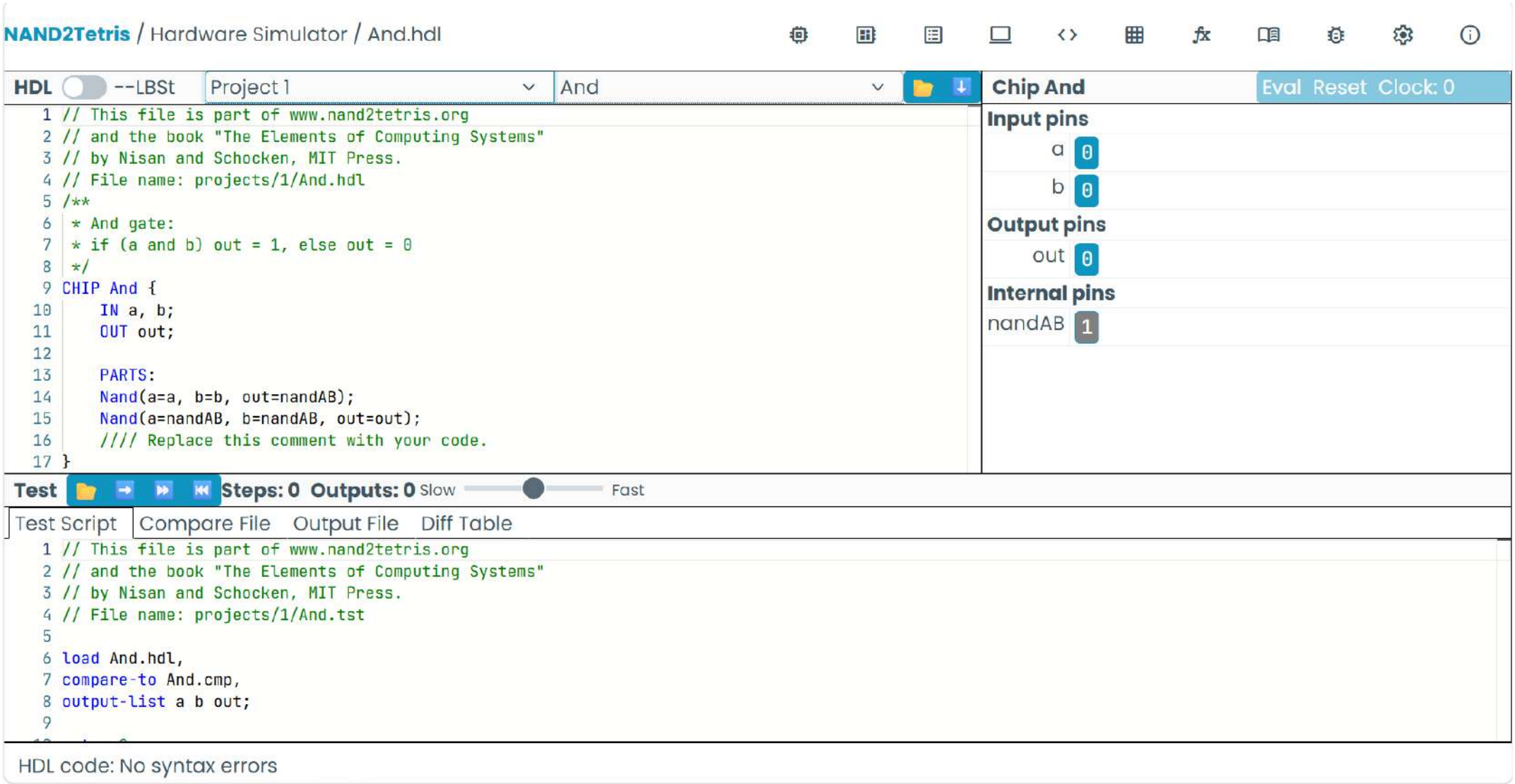
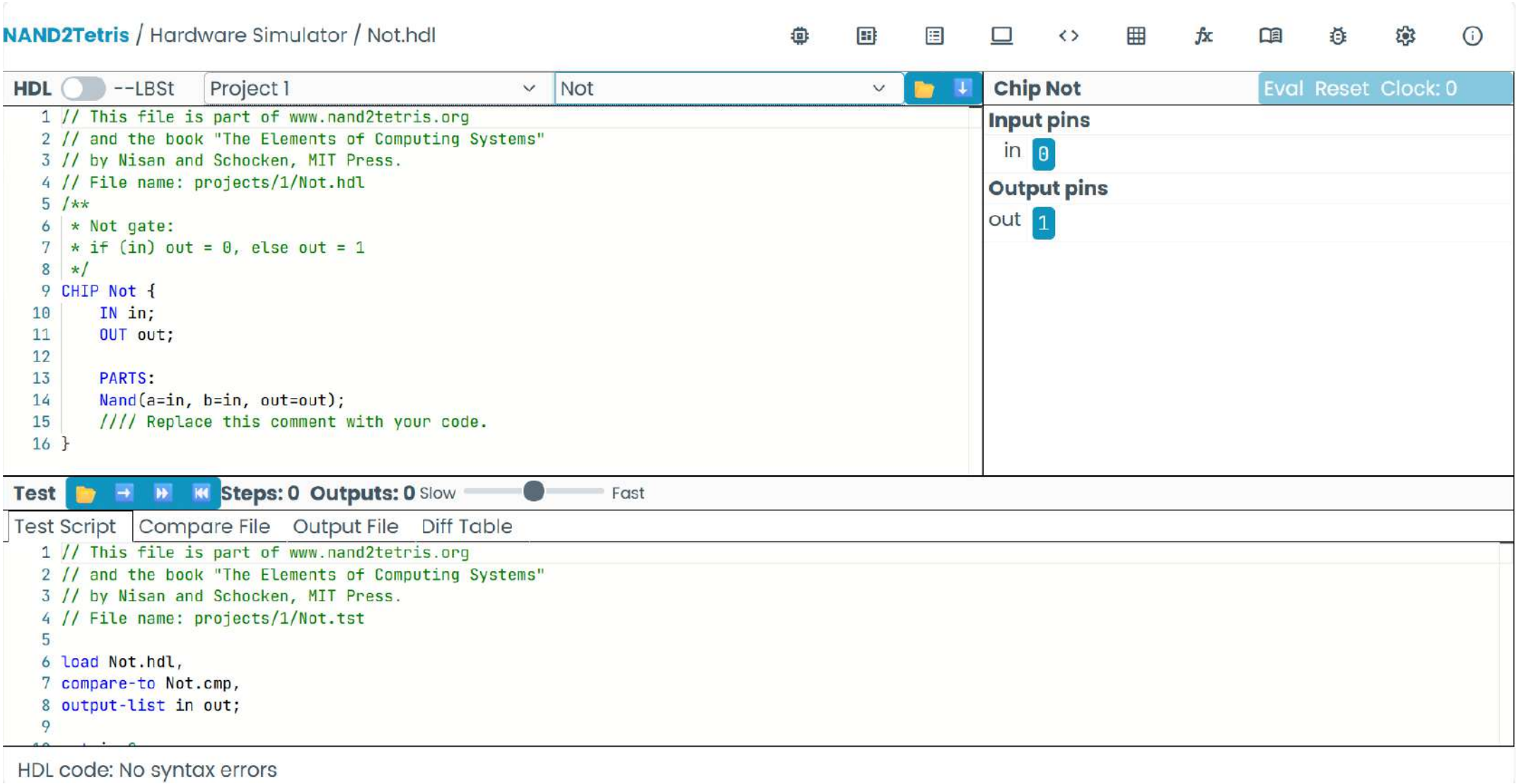
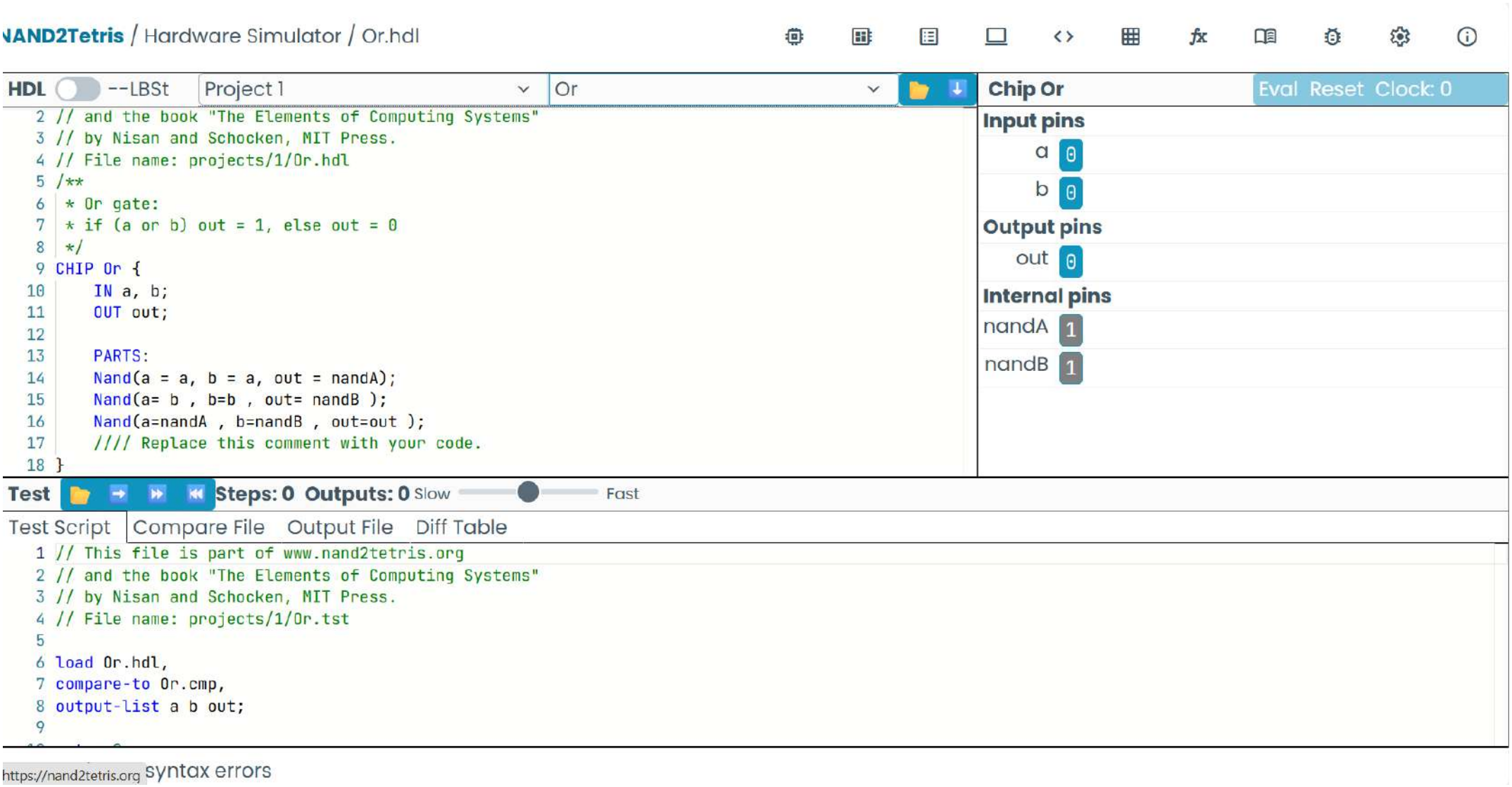
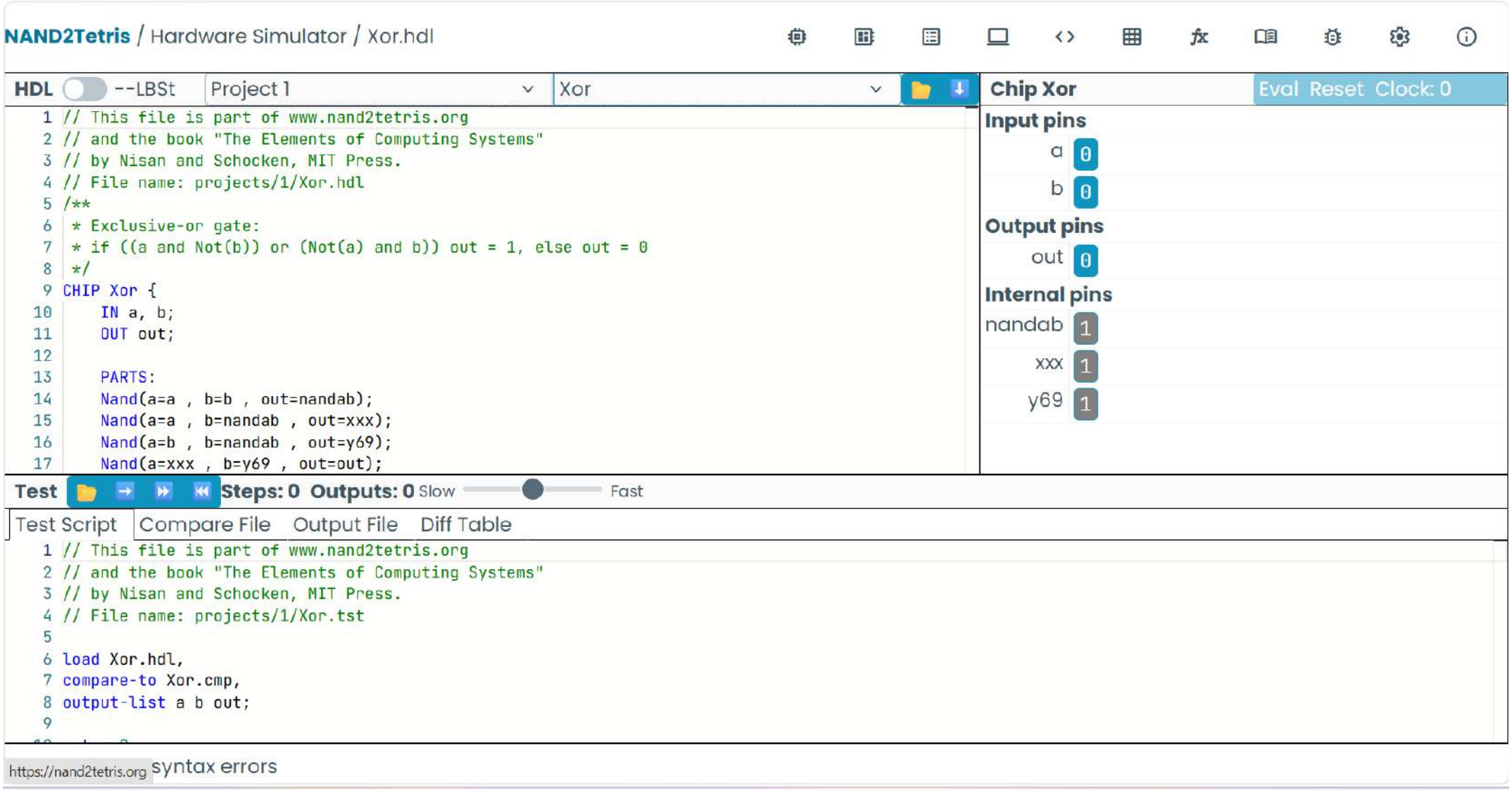
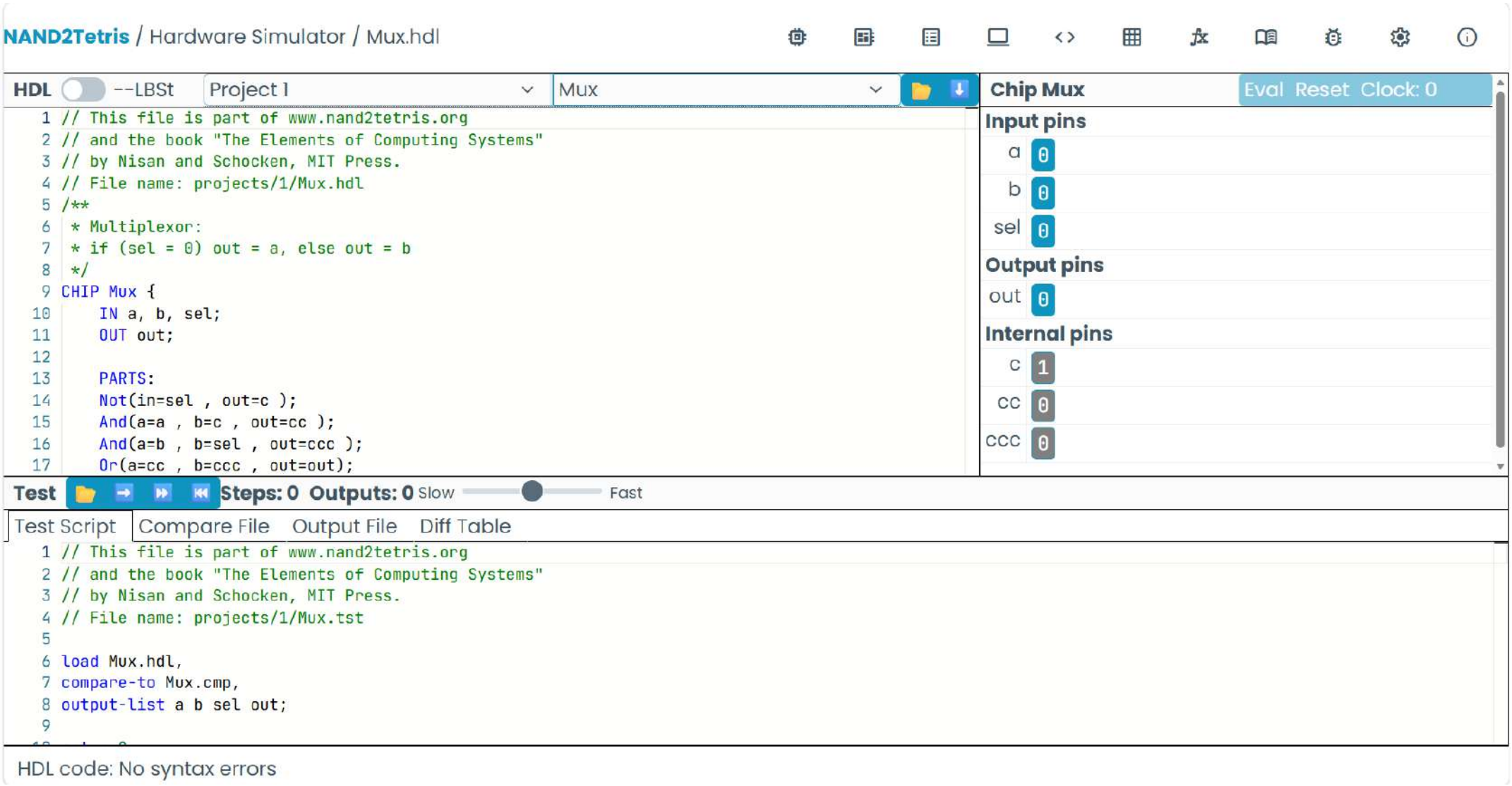
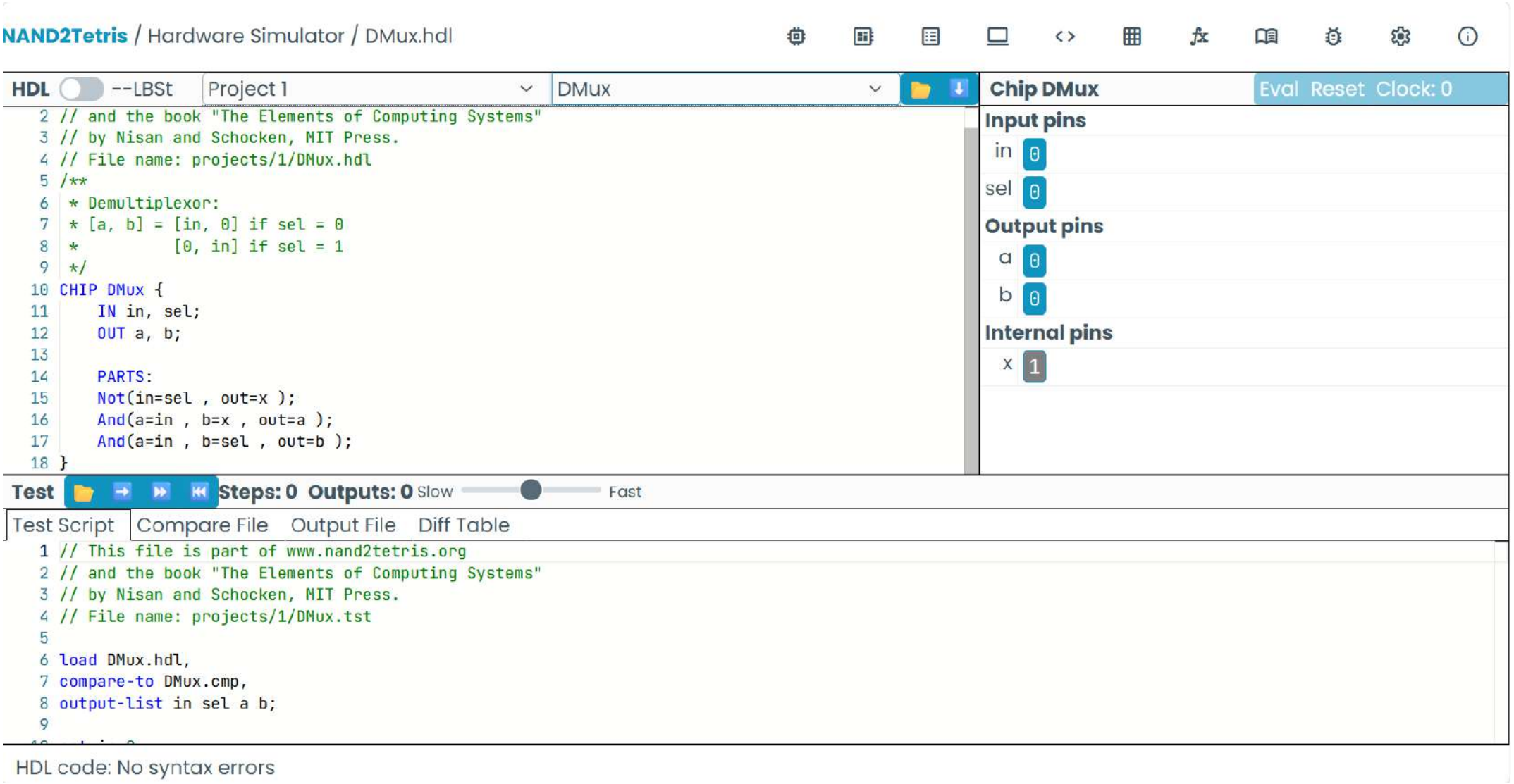

NAND2Tetris / Hardware Simulator / And.hdl
HDL --LBSt Project 1 And
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.hdl
5 /**
6 * And gate:
7 * if (a and b) out = 1, else out = 0
8 */
9 CHIP And {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a, b=b, out=nandAB);
15 Nand(a=nandAB, b=nandAB, out=out);
16 //// Replace this comment with your code.
17 }
Chip And
Eval Reset Clock: 0
Input pins
a 0
b 0
Output pins
out 0
Internal pins
nandAB 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.tst
5
6 load And.hdl,
7 compare-to And.cmp,
8 output-list a b out;
9
HDL code: No syntax errors

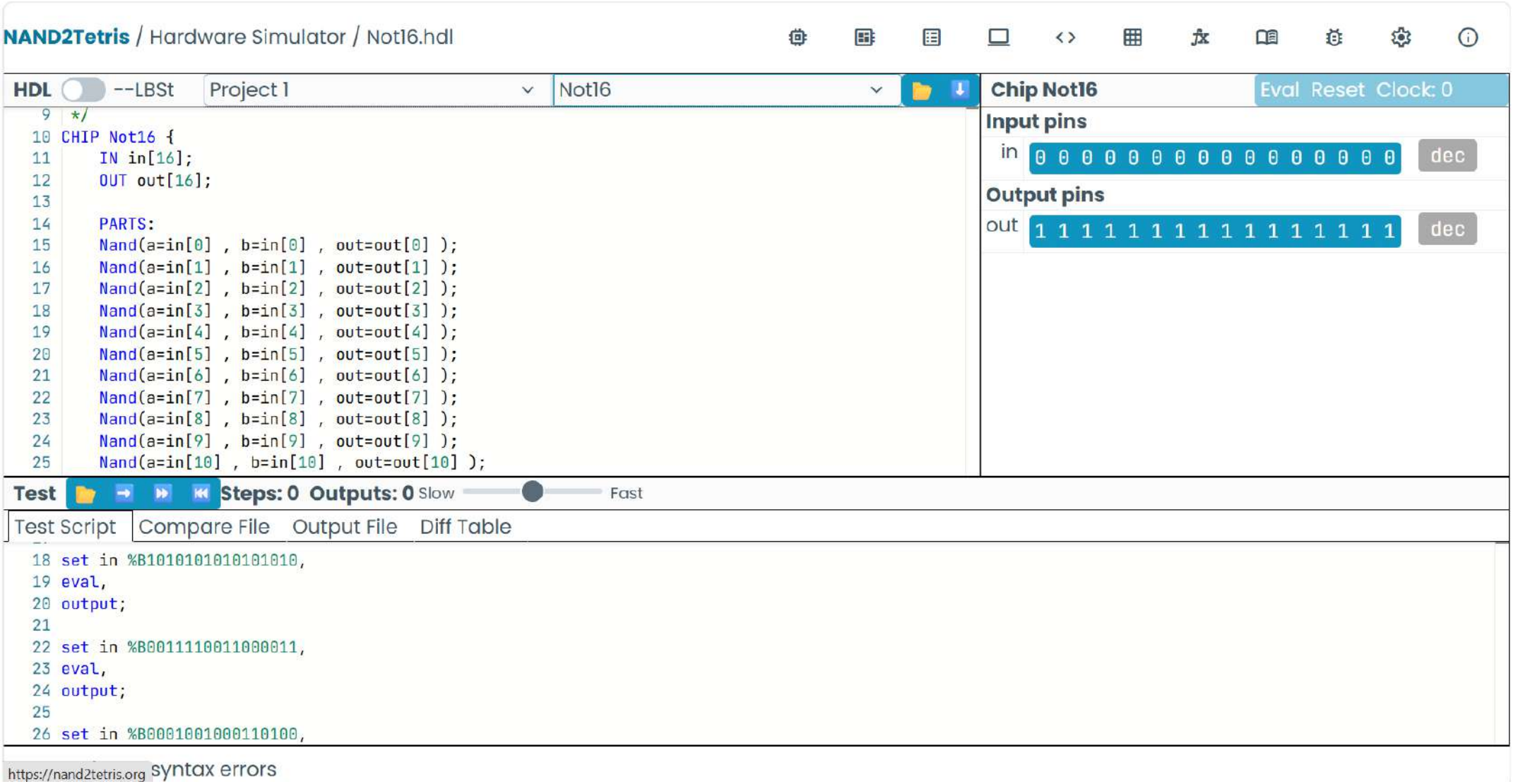

NAND2Tetris / Hardware Simulator / Not.hdl
HDL
--LBSt
Project 1
Not
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.hdl
5 /**
6 * Not gate:
7 * if (in) out = 0, else out = 1
8 */
9 CHIP Not {
10 IN in;
11 OUT out;
12
13 PARTS:
14 Nand(a=in, b=in, out=out);
15 //// Replace this comment with your code.
16 }
Chip Not
Eval Reset Clock: 0
Input pins
in 0
Output pins
out 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.tst
5
6 load Not.hdl,
7 compare-to Not.cmp,
8 output-list in out;
9
HDL code: No syntax errors

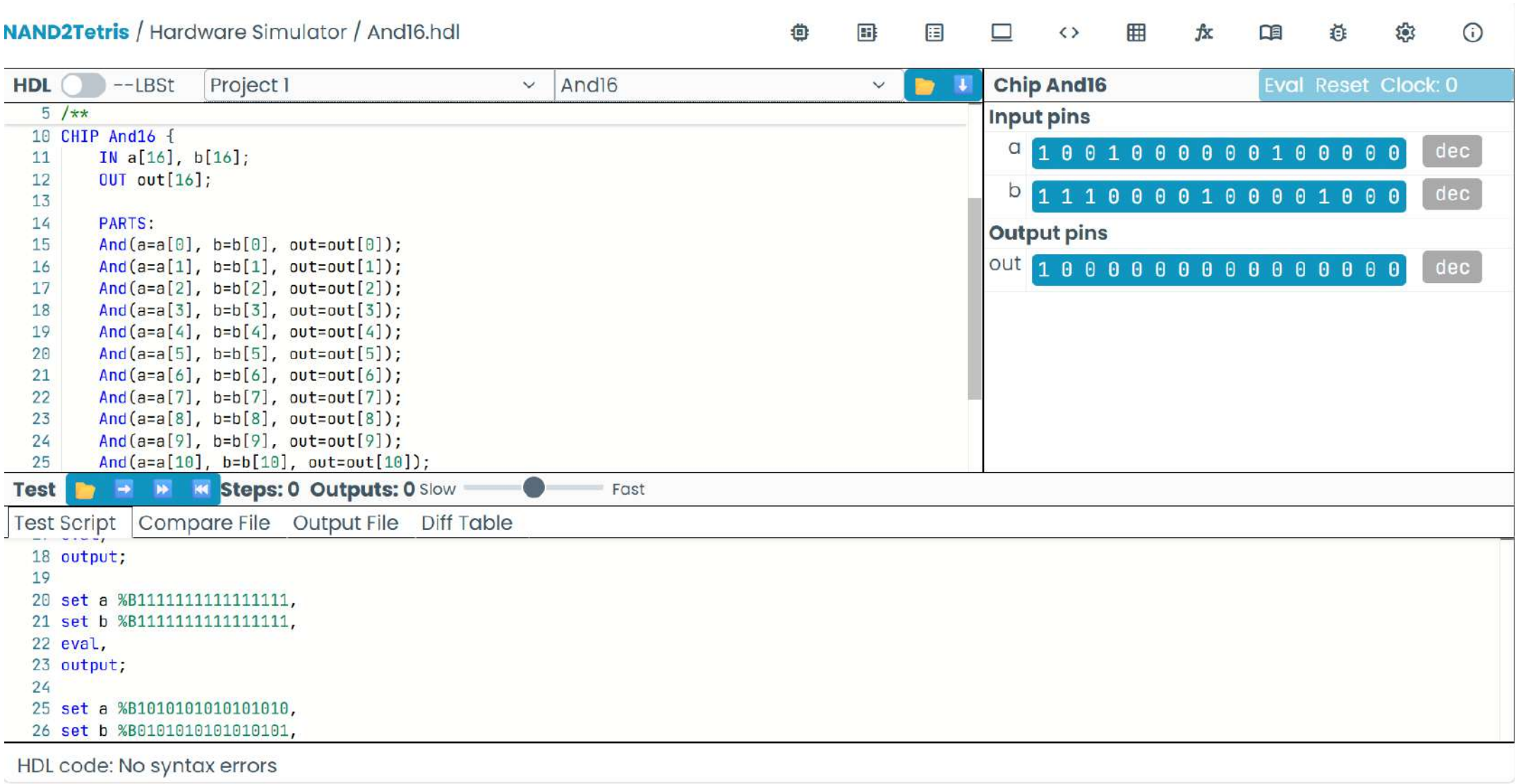

NAND2Tetris / Hardware Simulator / Or.hdl
HDL --LBSt
Project 1
Or
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or.hdl
/**
* Or gate:
* if (a or b) out = 1, else out = 0
*/
CHIP Or {
IN a, b;
OUT out;
PARTS:
Nand(a = a, b = a, out = nandA);
Nand(a= b , b=b , out= nandB);
Nand(a=nandA , b=nandB , out=out);
//// Replace this comment with your code.
}
Chip Or
Eval Reset Clock: 0
Input pins
a 0
b 0
Output pins
out 0
Internal pins
nandA 1
nandB 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or.tst
load Or.hdl,
compare-to Or.cmp,
output-list a b out;
https://nand2tetris.org
syntax errors

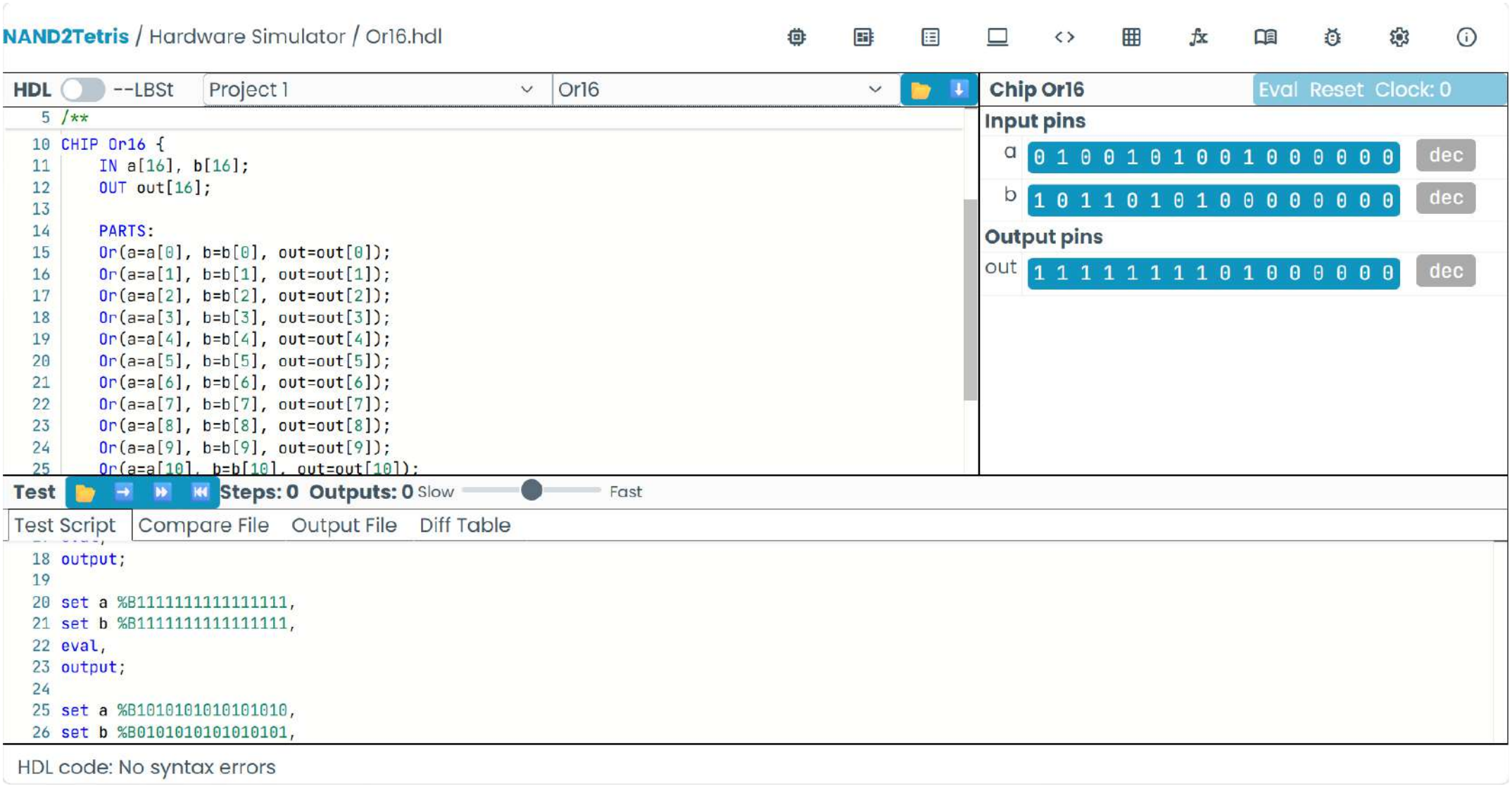

NAND2Tetris / Hardware Simulator / Xor.hdl
HDL --LBSt
Project 1
Xor
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Xor.hdl
5 /**
6 * Exclusive-or gate:
7 * if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
8 */
9 CHIP Xor {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a , b=b , out=nandab);
15 Nand(a=a , b=nandab , out=xxx);
16 Nand(a=b , b=nandab , out=y69);
17 Nand(a=xxx , b=y69 , out=out);
Chip Xor
Eval Reset Clock: 0
Input pins
a 0
b 0
Output pins
out 0
Internal pins
nandab 1
xxx 1
y69 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Xor.tst
5
6 load Xor.hdl,
7 compare-to Xor.cmp,
8 output-list a b out;
9
https://nand2tetris.org
syntax errors

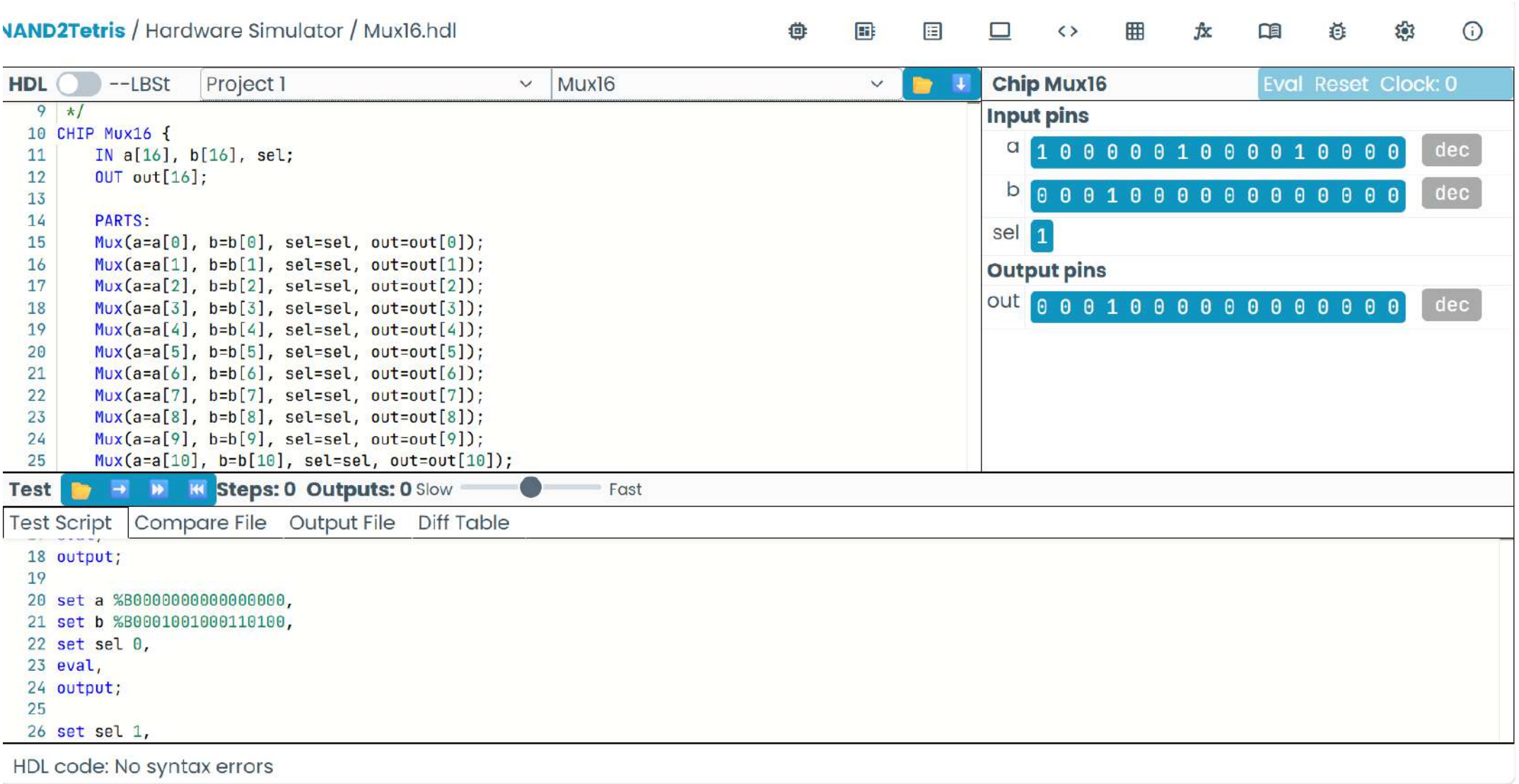

NAND2Tetris / Hardware Simulator / Mux.hdl
HDL --LBSt
Project 1
Mux
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux.hdl
5 /**
6 * Multiplexor:
7 * if (sel = 0) out = a, else out = b
8 */
9 CHIP Mux {
10 IN a, b, sel;
11 OUT out;
12
13 PARTS:
14 Not(in=sel , out=c);
15 And(a=a , b=c , out=cc);
16 And(a=b , b=sel , out=ccc);
17 Or(a=cc , b=ccc , out=out);
Chip Mux
Eval Reset Clock: 0
Input pins
a 0
b 0
sel 0
Output pins
out 0
Internal pins
c 1
cc 0
ccc 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux.tst
5
6 load Mux.hdl,
7 compare-to Mux.cmp,
8 output-list a b sel out;
9
HDL code: No syntax errors

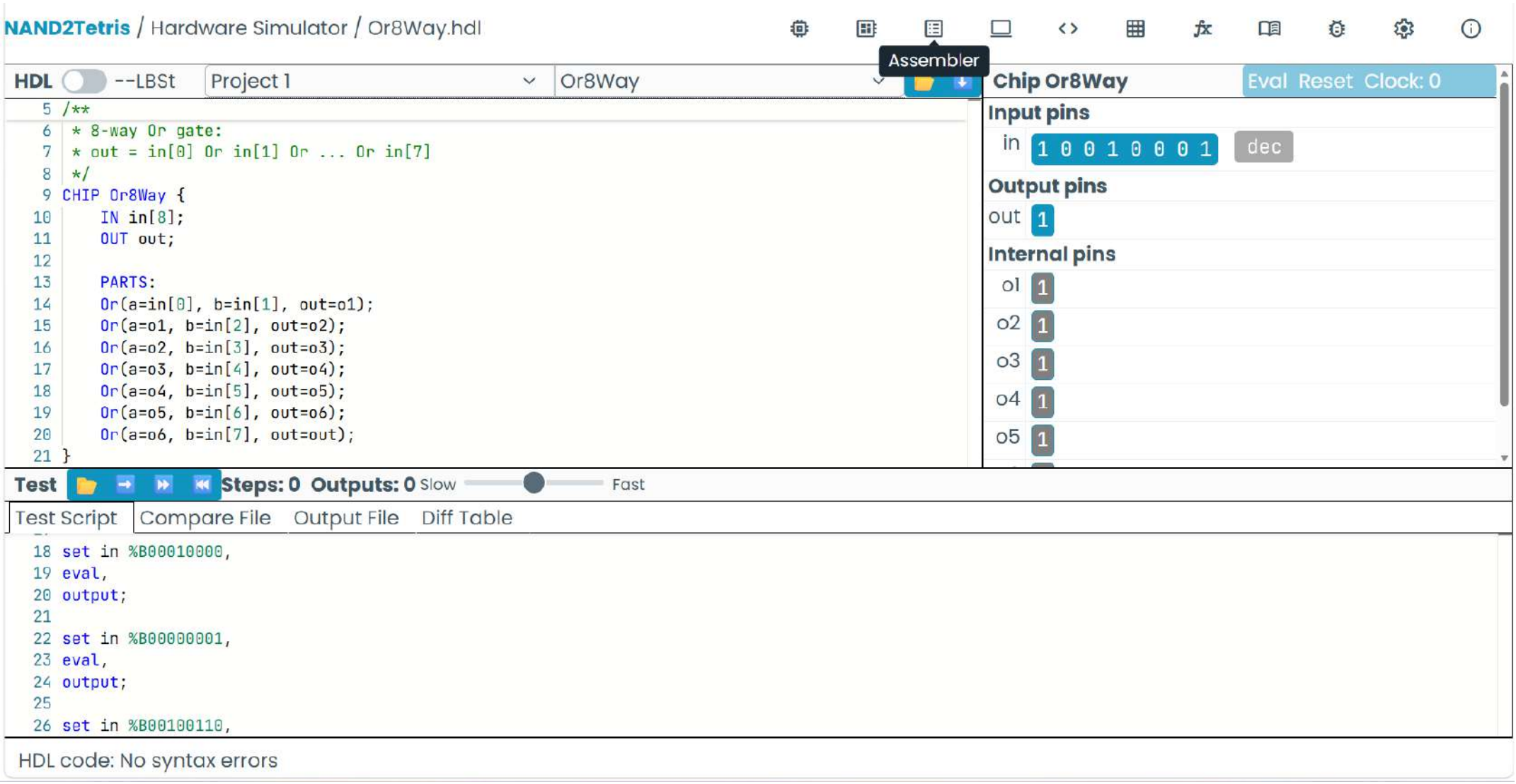

NAND2Tetris / Hardware Simulator / DMux.hdl
HDL --LBSt
Project 1
DMux
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux.hdl
/**
* Demultiplexor:
* [a, b] = [in, 0] if sel = 0
* [0, in] if sel = 1
*/
CHIP DMux {
IN in, sel;
OUT a, b;
PARTS:
Not(in=sel , out=x);
And(a=in , b=x , out=a);
And(a=in , b=sel , out=b);
}
Chip DMux
Eval Reset Clock: 0
Input pins
in 0
sel 0
Output pins
a 0
b 0
Internal pins
x 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux.tst
load DMux.hdl,
compare-to DMux.cmp,
output-list in sel a b;
HDL code: No syntax errors

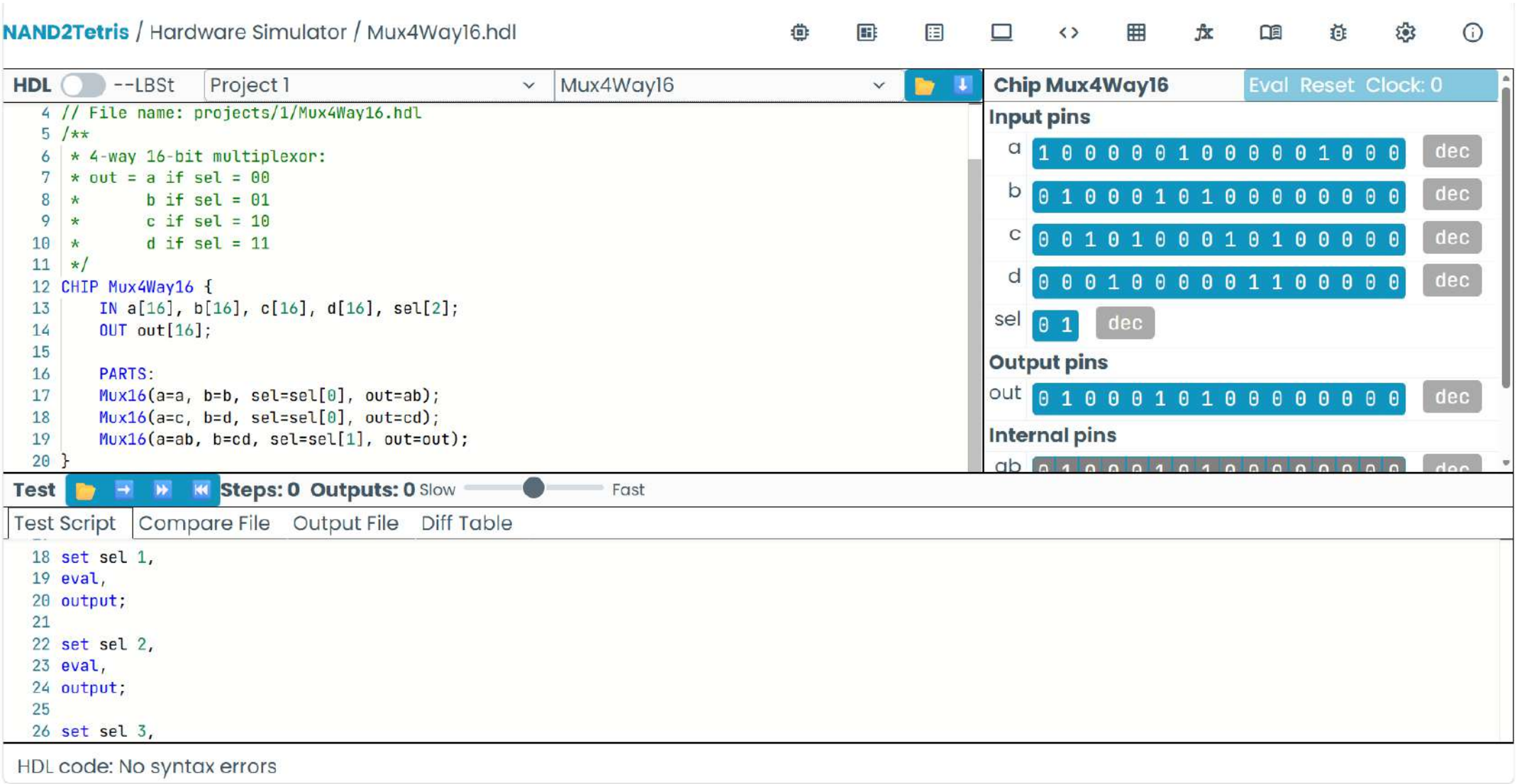

NAND2Tetris / Hardware Simulator / Not16.hdl
HDL --LBSt Project 1 Not16
9 */
10 CHIP Not16 {
11 IN in[16];
12 OUT out[16];
13
14 PARTS:
15 Nand(a=in[0] , b=in[0] , out=out[0]);
16 Nand(a=in[1] , b=in[1] , out=out[1]);
17 Nand(a=in[2] , b=in[2] , out=out[2]);
18 Nand(a=in[3] , b=in[3] , out=out[3]);
19 Nand(a=in[4] , b=in[4] , out=out[4]);
20 Nand(a=in[5] , b=in[5] , out=out[5]);
21 Nand(a=in[6] , b=in[6] , out=out[6]);
22 Nand(a=in[7] , b=in[7] , out=out[7]);
23 Nand(a=in[8] , b=in[8] , out=out[8]);
24 Nand(a=in[9] , b=in[9] , out=out[9]);
25 Nand(a=in[10] , b=in[10] , out=out[10]);
Chip Not16
Eval Reset Clock: 0
Input pins
in 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Output pins
out 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
18 set in %B1010101010101010,
19 eval,
20 output;
21
22 set in %B0011110011000011,
23 eval,
24 output;
25
26 set in %B0001001000110100,
https://nand2tetris.org syntax errors

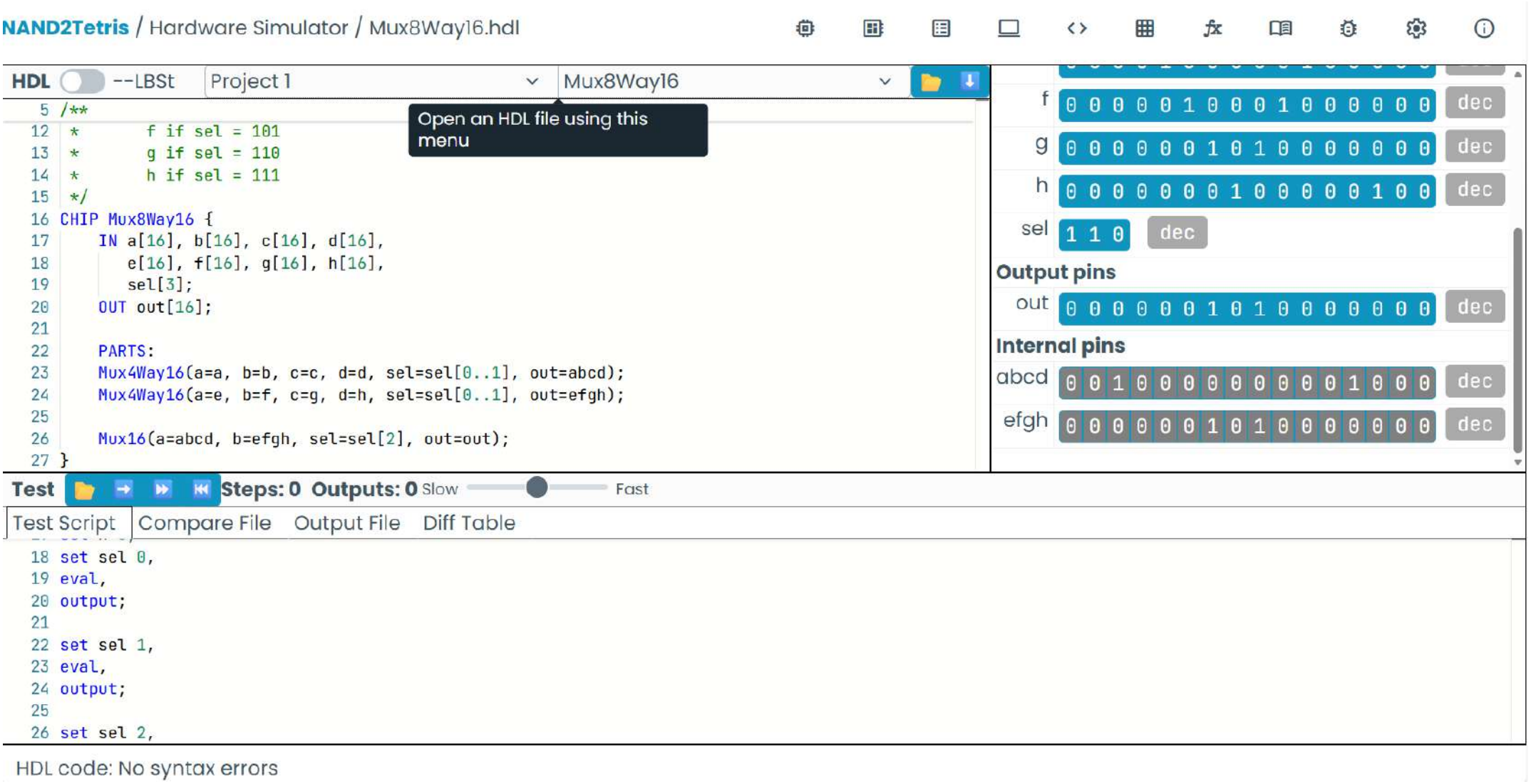

NAND2Tetris / Hardware Simulator / And16.hdl
HDL --LBSt Project 1 And16
5 /**
10 CHIP And16 {
11 IN a[16], b[16];
12 OUT out[16];
13
14 PARTS:
15 And(a=a[0], b=b[0], out=out[0]);
16 And(a=a[1], b=b[1], out=out[1]);
17 And(a=a[2], b=b[2], out=out[2]);
18 And(a=a[3], b=b[3], out=out[3]);
19 And(a=a[4], b=b[4], out=out[4]);
20 And(a=a[5], b=b[5], out=out[5]);
21 And(a=a[6], b=b[6], out=out[6]);
22 And(a=a[7], b=b[7], out=out[7]);
23 And(a=a[8], b=b[8], out=out[8]);
24 And(a=a[9], b=b[9], out=out[9]);
25 And(a=a[10], b=b[10], out=out[10]);
Chip And16
Eval Reset Clock: 0
Input pins
a 1 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 dec
b 1 1 1 0 0 0 0 1 0 0 0 0 1 0 0 0 dec
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
18 output;
19
20 set a %B1111111111111111,
21 set b %B1111111111111111,
22 eval,
23 output;
24
25 set a %B1010101010101010,
26 set b %B0101010101010101,
HDL code: No syntax errors

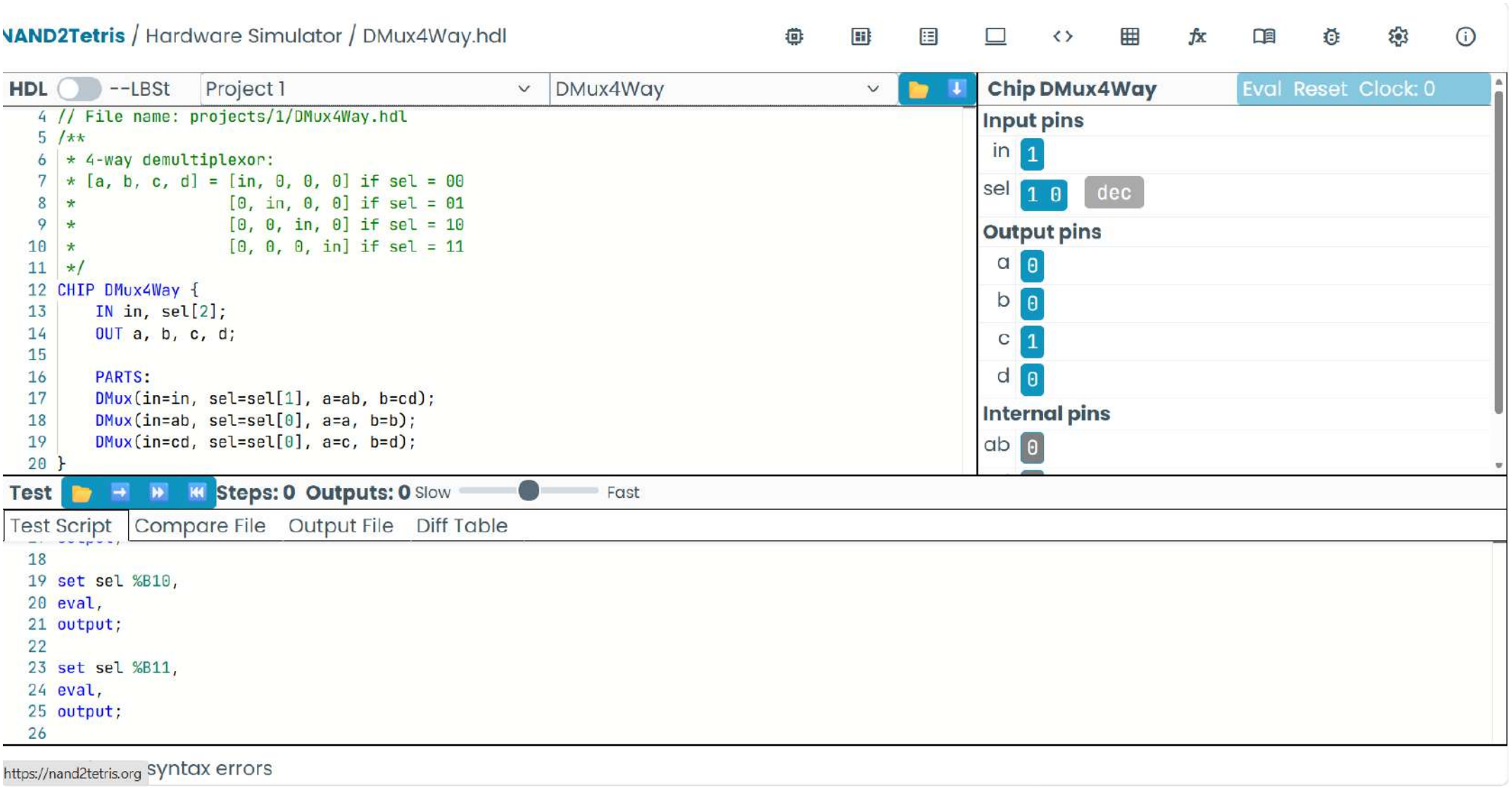

NAND2Tetris / Hardware Simulator / Or16.hdl
HDL --LBSt
Project 1
Or16
/**
CHIP Or16 {
IN a[16], b[16];
OUT out[16];
PARTS:
Or(a=a[0], b=b[0], out=out[0]);
Or(a=a[1], b=b[1], out=out[1]);
Or(a=a[2], b=b[2], out=out[2]);
Or(a=a[3], b=b[3], out=out[3]);
Or(a=a[4], b=b[4], out=out[4]);
Or(a=a[5], b=b[5], out=out[5]);
Or(a=a[6], b=b[6], out=out[6]);
Or(a=a[7], b=b[7], out=out[7]);
Or(a=a[8], b=b[8], out=out[8]);
Or(a=a[9], b=b[9], out=out[9]);
Or(a=a[10], b=b[10], out=out[10]);
Chip Or16
Eval Reset Clock: 0
Input pins
a 0 1 0 0 1 0 1 0 0 1 0 0 0 0 0 0 dec
b 1 0 1 1 0 1 0 1 0 0 0 0 0 0 0 0 dec
Output pins
out 1 1 1 1 1 1 1 1 0 1 0 0 0 0 0 0 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
HDL code: No syntax errors

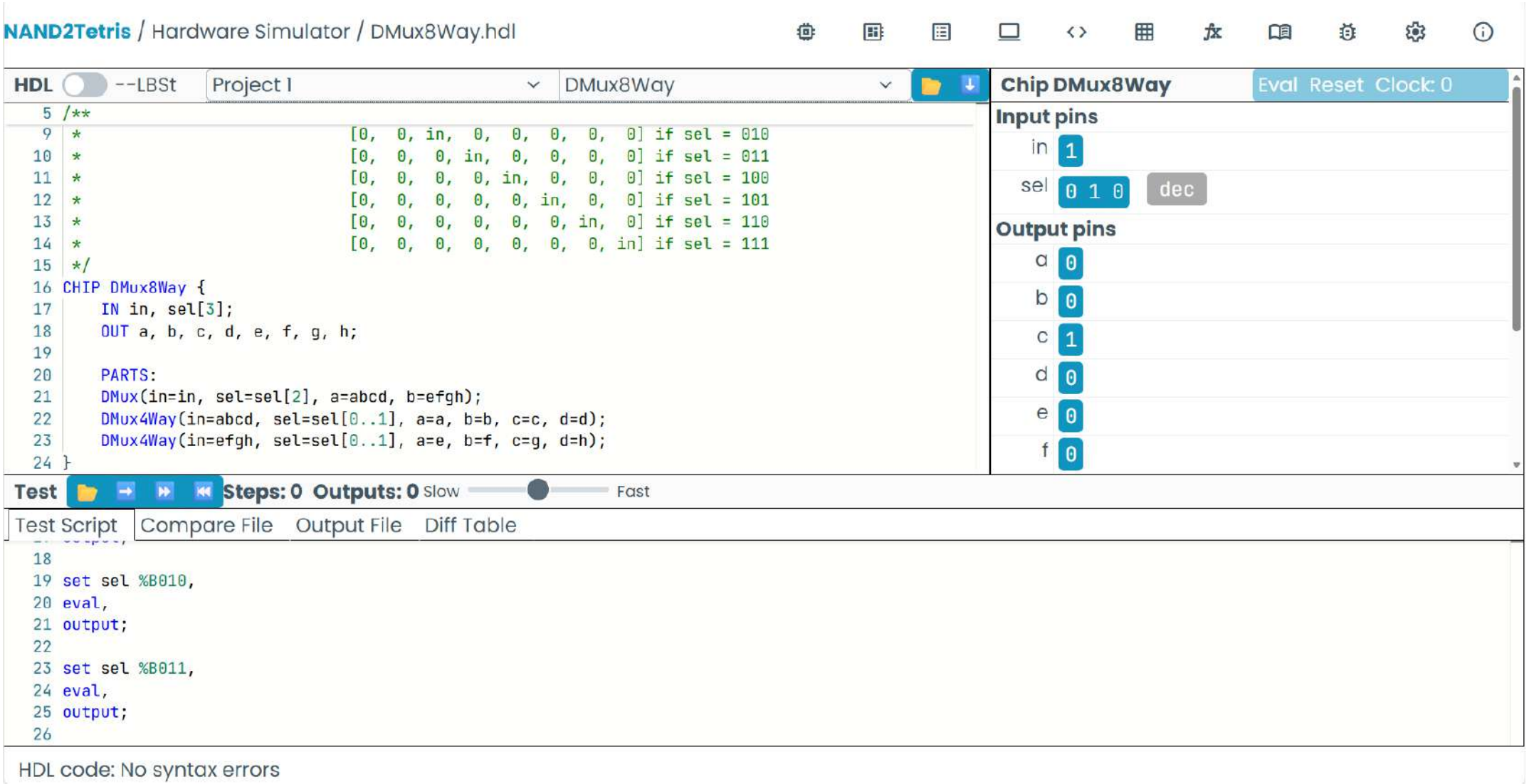

NAND2Tetris / Hardware Simulator / Mux16.hdl
HDL --LBSt Project 1 Mux16
*/
CHIP Mux16 {
IN a[16], b[16], sel;
OUT out[16];
PARTS:
Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
Chip Mux16 Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 dec
b 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 dec
sel 1
Output pins
out 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
output;
set a %B0000000000000000,
set b %B0001001000110100,
set sel 0,
eval,
output;
set sel 1,
HDL code: No syntax errors

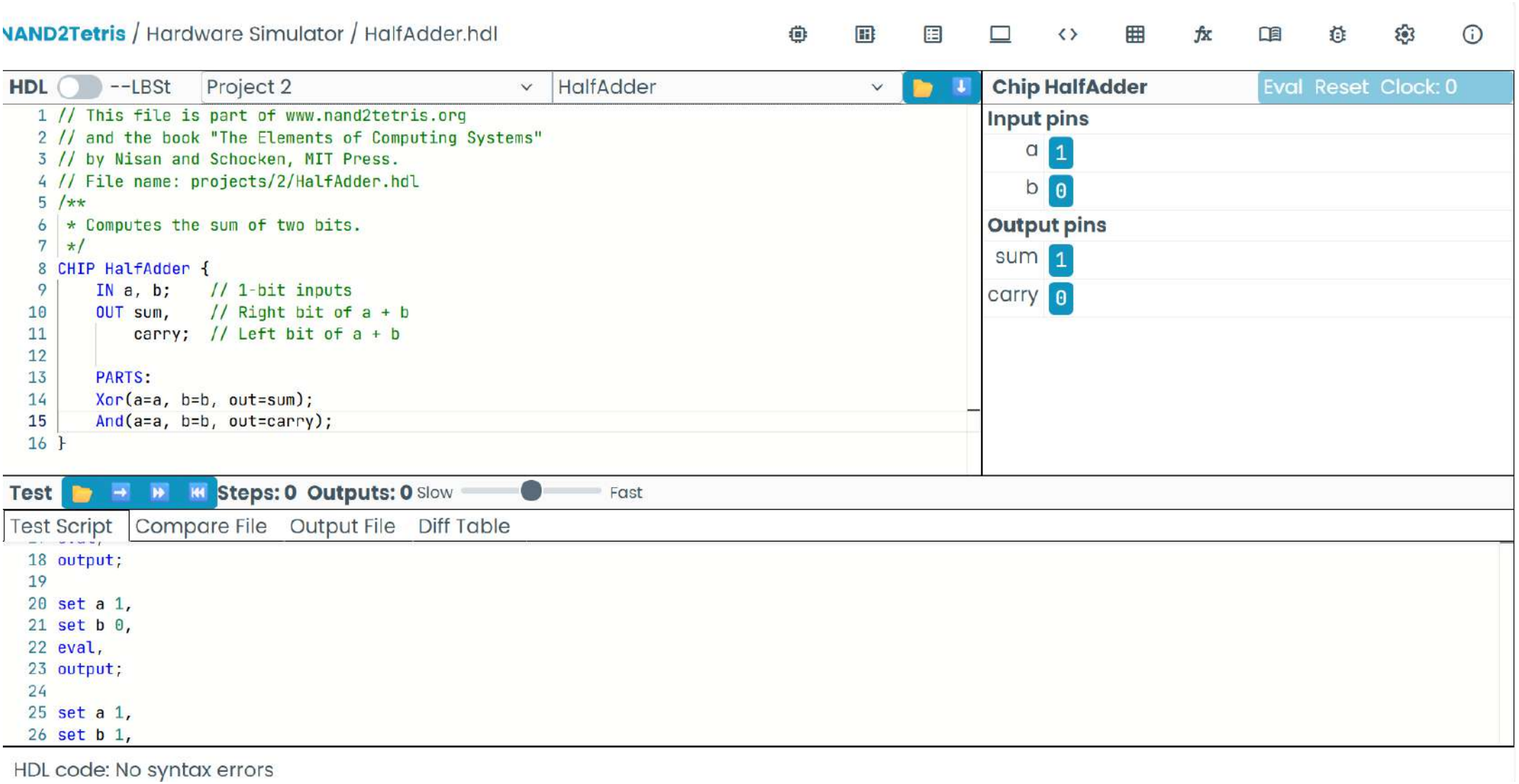

NAND2Tetris / Hardware Simulator / Or8Way.hdl
Assembler
HDL --LBSt
Project 1
Or8Way
/**
* 8-way Or gate:
* out = in[0] Or in[1] Or ... Or in[7]
*/
CHIP Or8Way {
IN in[8];
OUT out;
PARTS:
Or(a=in[0], b=in[1], out=o1);
Or(a=o1, b=in[2], out=o2);
Or(a=o2, b=in[3], out=o3);
Or(a=o3, b=in[4], out=o4);
Or(a=o4, b=in[5], out=o5);
Or(a=o5, b=in[6], out=o6);
Or(a=o6, b=in[7], out=out);
}
Chip Or8Way
Eval Reset Clock: 0
Input pins
in 1 0 0 1 0 0 0 1
dec
Output pins
out 1
Internal pins
o1 1
o2 1
o3 1
o4 1
o5 1
Test
Steps: 0 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set in %B00010000,
eval,
output;
set in %B00000001,
eval,
output;
set in %B00100110,
HDL code: No syntax errors

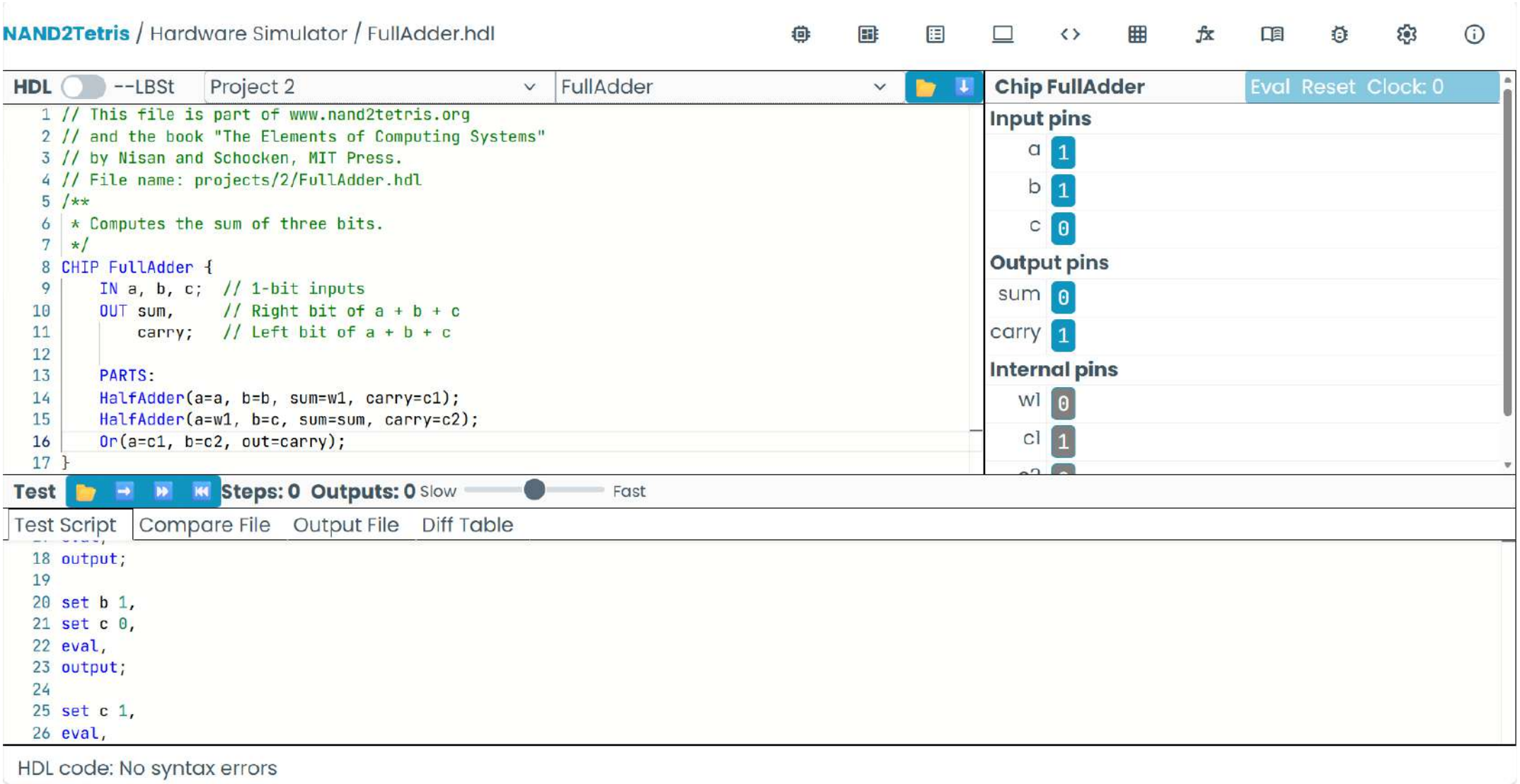

NAND2Tetris / Hardware Simulator / Mux4Way16.hdl
HDL --LBSt
Project 1
Mux4Way16
4 // File name: projects/1/Mux4Way16.hdl
5 /**
6 * 4-way 16-bit multiplexor:
7 * out = a if sel = 00
8 * b if sel = 01
9 * c if sel = 10
10 * d if sel = 11
11 */
12 CHIP Mux4Way16 {
13 IN a[16], b[16], c[16], d[16], sel[2];
14 OUT out[16];
15
16 PARTS:
17 Mux16(a=a, b=b, sel=sel[0], out=ab);
18 Mux16(a=c, b=d, sel=sel[0], out=cd);
19 Mux16(a=ab, b=cd, sel=sel[1], out=out);
20 }
Chip Mux4Way16
Eval Reset Clock: 0
Input pins
a 1 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 dec
b 0 1 0 0 0 1 0 1 0 0 0 0 0 0 0 0 dec
c 0 0 1 0 1 0 0 0 1 0 1 0 0 0 0 0 dec
d 0 0 0 1 0 0 0 0 0 1 1 0 0 0 0 0 dec
sel 0 1 dec
Output pins
out 0 1 0 0 0 1 0 1 0 0 0 0 0 0 0 0 dec
Internal pins
ab
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
18 set sel 1,
19 eval,
20 output;
21
22 set sel 2,
23 eval,
24 output;
25
26 set sel 3,
HDL code: No syntax errors

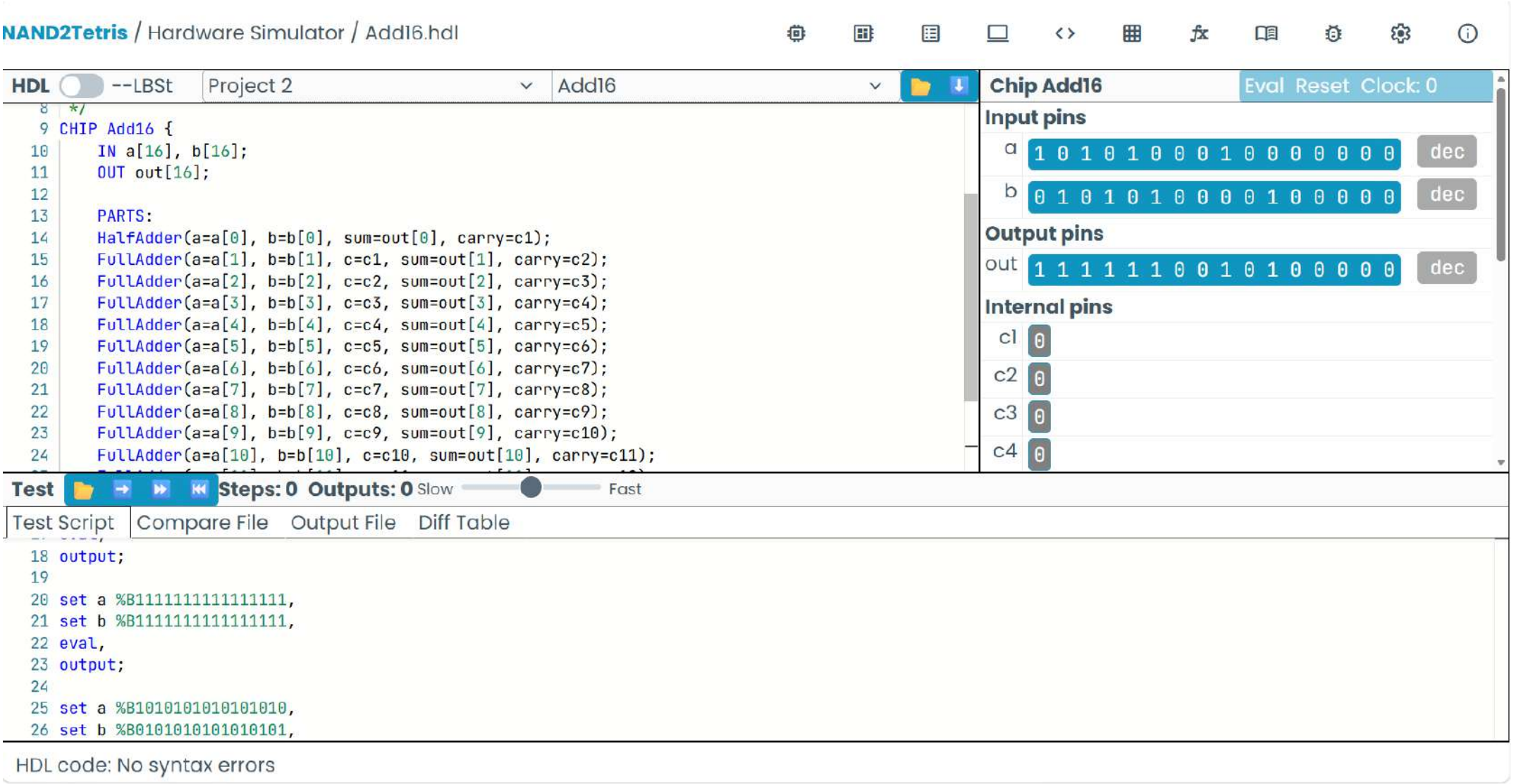

NAND2Tetris / Hardware Simulator / Mux8Way16.hdl
HDL
--LBSt
Project 1
Mux8Way16
Open an HDL file using this menu
5 /**
12 * f if sel = 101
13 * g if sel = 110
14 * h if sel = 111
15 */
16 CHIP Mux8Way16 {
17 IN a[16], b[16], c[16], d[16],
18 e[16], f[16], g[16], h[16],
19 sel[3];
20 OUT out[16];
21
22 PARTS:
23 Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=abcd);
24 Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=efgh);
25
26 Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
27 }
f 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 dec
g 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 dec
h 0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 dec
sel 1 1 0 dec
Output pins
out 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 dec
Internal pins
abcd 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 dec
efgh 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
18 set sel 0,
19 eval,
20 output;
21
22 set sel 1,
23 eval,
24 output;
25
26 set sel 2,
HDL code: No syntax errors

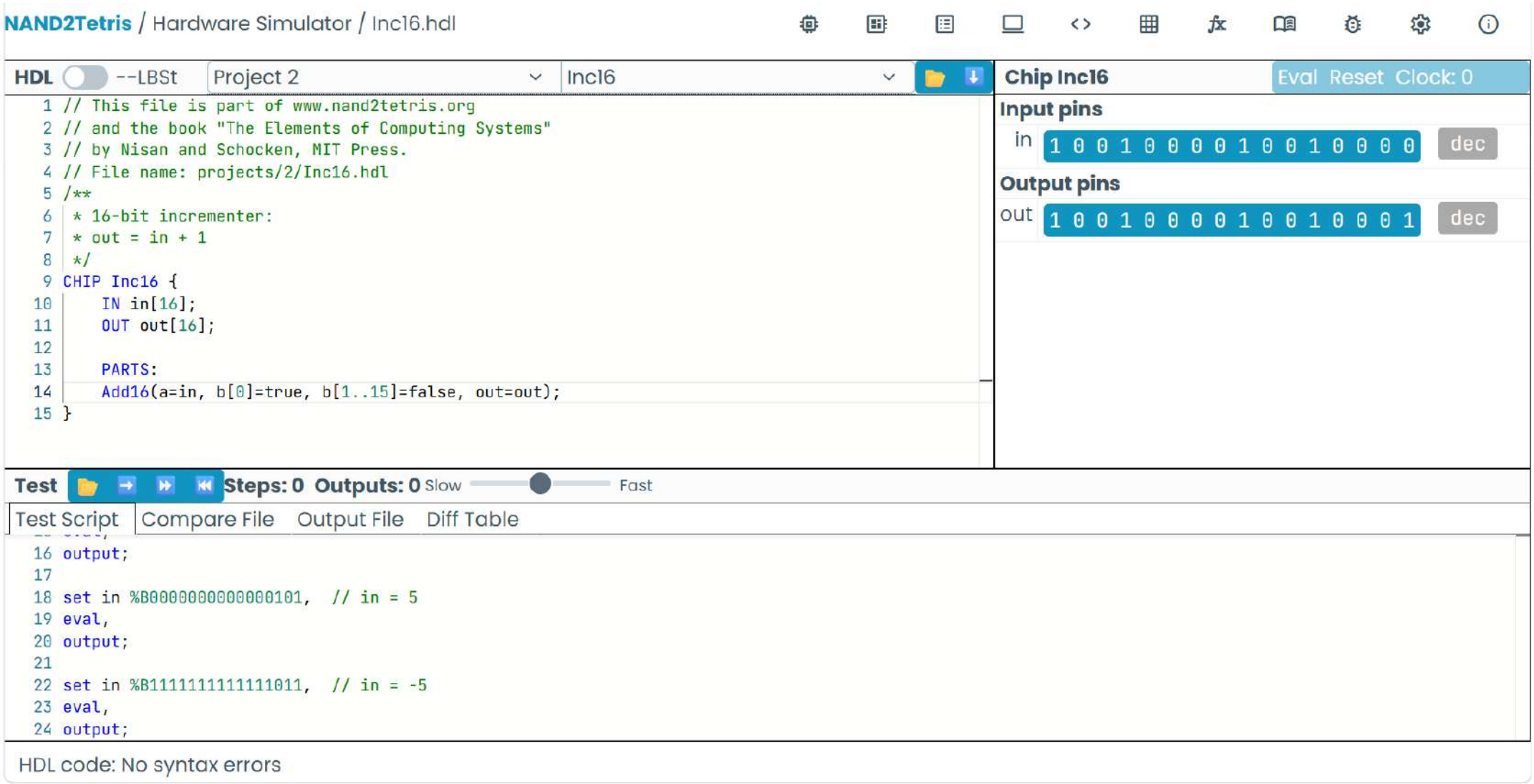

NAND2Tetris / Hardware Simulator / DMux4Way.hdl
HDL --LBSt
Project 1
DMux4Way
4 // File name: projects/1/DMux4Way.hdl
5 /**
6 * 4-way demultiplexor:
7 * [a, b, c, d] = [in, 0, 0, 0] if sel = 00
8 * [0, in, 0, 0] if sel = 01
9 * [0, 0, in, 0] if sel = 10
10 * [0, 0, 0, in] if sel = 11
11 */
12 CHIP DMux4Way {
13 IN in, sel[2];
14 OUT a, b, c, d;
15
16 PARTS:
17 DMux(in=in, sel=sel[1], a=ab, b=cd);
18 DMux(in=ab, sel=sel[0], a=a, b=b);
19 DMux(in=cd, sel=sel[0], a=c, b=d);
20 }
Chip DMux4Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 0 dec
Output pins
a 0
b 0
c 1
d 0
Internal pins
ab 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
18
19 set sel %B10,
20 eval,
21 output;
22
23 set sel %B11,
24 eval,
25 output;
26
https://nand2tetris.org
syntax errors

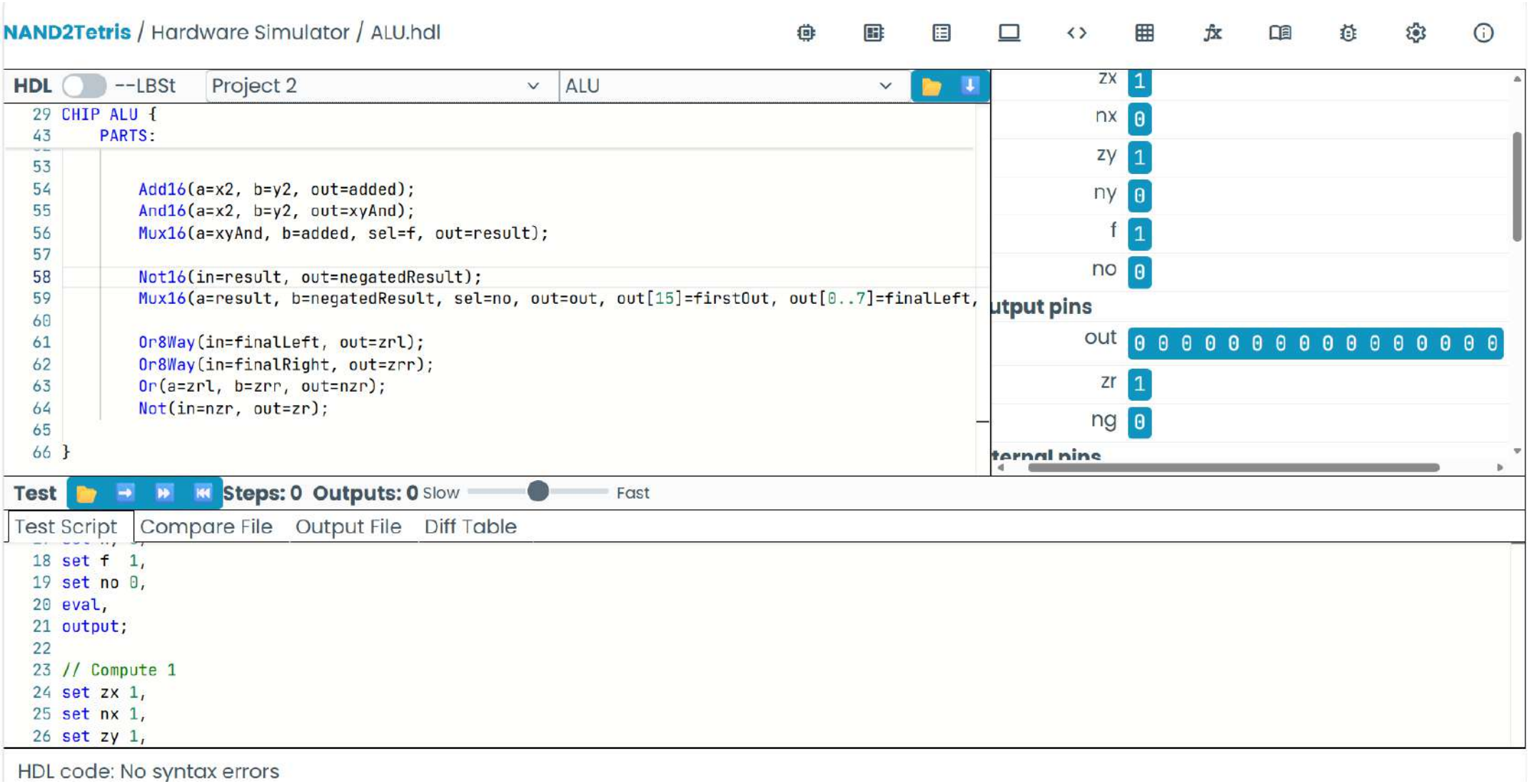

NAND2Tetris / Hardware Simulator / DMux8Way.hdl
HDL --LBSt
Project 1
DMux8Way
/**
* [0, 0, in, 0, 0, 0, 0, 0] if sel = 010
* [0, 0, 0, in, 0, 0, 0, 0] if sel = 011
* [0, 0, 0, 0, in, 0, 0, 0] if sel = 100
* [0, 0, 0, 0, 0, in, 0, 0] if sel = 101
* [0, 0, 0, 0, 0, 0, in, 0] if sel = 110
* [0, 0, 0, 0, 0, 0, 0, in] if sel = 111
*/
CHIP DMux8Way {
IN in, sel[3];
OUT a, b, c, d, e, f, g, h;
PARTS:
DMux(in=in, sel=sel[2], a=abcd, b=efgh);
DMux4Way(in=abcd, sel=sel[0..1], a=a, b=b, c=c, d=d);
DMux4Way(in=efgh, sel=sel[0..1], a=e, b=f, c=g, d=h);
}
Chip DMux8Way
Eval Reset Clock: 0
Input pins
in 1
sel 0 1 0 dec
Output pins
a 0
b 0
c 1
d 0
e 0
f 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
set sel %B010,
eval,
output;
set sel %B011,
eval,
output;
HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt
Project 2
HalfAdder
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/HalfAdder.hdl
5 /**
6 * Computes the sum of two bits.
7 */
8 CHIP HalfAdder {
9 IN a, b; // 1-bit inputs
10 OUT sum, // Right bit of a + b
11 carry; // Left bit of a + b
12
13 PARTS:
14 Xor(a=a, b=b, out=sum);
15 And(a=a, b=b, out=carry);
16 }
Chip HalfAdder
Eval Reset Clock: 0
Input pins
a 1
b 0
Output pins
sum 1
carry 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
18 output;
19
20 set a 1,
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt
Project 2
FullAdder
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.hdl
/**
* Computes the sum of three bits.
*/
CHIP FullAdder {
IN a, b, c; // 1-bit inputs
OUT sum, // Right bit of a + b + c
carry; // Left bit of a + b + c
PARTS:
HalfAdder(a=a, b=b, sum=w1, carry=c1);
HalfAdder(a=w1, b=c, sum=sum, carry=c2);
Or(a=c1, b=c2, out=carry);
}
Chip FullAdder
Eval Reset Clock: 0
Input pins
a 1
b 1
c 0
Output pins
sum 0
carry 1
Internal pins
w1 0
c1 1
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
output;
set b 1,
set c 0,
eval,
output;
set c 1,
eval,
HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt
Project 2
Add16
*/
CHIP Add16 {
IN a[16], b[16];
OUT out[16];
PARTS:
HalfAdder(a=a[0], b=b[0], sum=out[0], carry=c1);
FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=c2);
FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=c3);
FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=c4);
FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=c5);
FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=c7);
FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
Chip Add16
Eval Reset Clock: 0
Input pins
a 1 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 dec
b 0 1 0 1 0 1 0 0 0 0 1 0 0 0 0 0 dec
Output pins
out 1 1 1 1 1 1 0 0 1 0 1 0 0 0 0 0 dec
Internal pins
c1 0
c2 0
c3 0
c4 0
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / Inc16.hdl
HDL --LBSt Project 2 Inc16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Inc16.hdl
5 /**
6 * 16-bit incrementer:
7 * out = in + 1
8 */
9 CHIP Inc16 {
10 IN in[16];
11 OUT out[16];
12
13 PARTS:
14 Add16(a=in, b[0]=true, b[1..15]=false, out=out);
15 }
Chip Inc16 Eval Reset Clock: 0
Input pins
in 1 0 0 1 0 0 0 0 1 0 0 1 0 0 0 0 dec
Output pins
out 1 0 0 1 0 0 0 0 1 0 0 1 0 0 0 1 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
16 output;
17
18 set in %B0000000000000101, // in = 5
19 eval,
20 output;
21
22 set in %B1111111111111011, // in = -5
23 eval,
24 output;
HDL code: No syntax errors


NAND2Tetris / Hardware Simulator / ALU.hdl
HDL
--LBSt
Project 2
ALU
29 CHIP ALU {
43 PARTS:
54 Add16(a=x2, b=y2, out=added);
55 And16(a=x2, b=y2, out=xyAnd);
56 Mux16(a=xyAnd, b=added, sel=f, out=result);
58 Not16(in=result, out=negatedResult);
59 Mux16(a=result, b=negatedResult, sel=no, out=out, out[15]=firstOut, out[0..7]=finalLeft,
61 Or8Way(in=finalLeft, out=zrl);
62 Or8Way(in=finalRight, out=zrr);
63 Or(a=zrl, b=zrr, out=nzr);
64 Not(in=nzr, out=zr);
66 }
zx 1
nx 0
zy 1
ny 0
f 1
no 0
utput pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
zr 1
ng 0
Test
Steps: 0 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
18 set f 1,
19 set no 0,
20 eval,
21 output;
23 // Compute 1
24 set zx 1,
25 set nx 1,
26 set zy 1,
HDL code: No syntax errors